

07_PRODUCTION_READINESS_CHECKLIST

- [SUOP ERP v1.0 — Final Production Readiness Checklist](#)
 - [How to Use This Checklist](#)
 - [1. Documentation \(100% complete\)](#)
 - [2. Foundation Framework \(Phase 0 Code\)](#)
 - [3. Database](#)
 - [4. API](#)
 - [5. Security](#)
 - [6. DevOps](#)
 - [7. Performance](#)
 - [8. Disaster Recovery](#)
 - [9. Monitoring & Observability](#)
 - [10. Testing](#)
 - [11. First Business Journey \(Procure-to-Receive\)](#)
 - [12. Production Launch Readiness](#)
 - [Final Production Readiness Score](#)
 - [Approval Block](#)

SUOP ERP v1.0 — Final Production Readiness Checklist

Field	Value
Document Version	1.0
Status	DRAFT — Awaiting Final Approval
Date	2026-07-11
Approval Required	Project Owner
Purpose	Single source of truth for v1.0 production readiness

How to Use This Checklist

This checklist defines **100% completion criteria** for SUOP ERP v1.0 production readiness. When every item is checked, the documentation phase is **permanently frozen** and implementation may begin.

The checklist aggregates requirements from all 8 architecture documents: - Phase 0 Architecture - Data Architecture - API Standards - Security Architecture - DevOps Architecture - Performance Standards - Disaster Recovery - (Plus operational items)

Each checkbox must be verifiable. “Verifiable” means there is a test, command, or artifact that proves the item is complete.

1. Documentation (100% complete)

#	Item	Verified By	Status
1.1	Phase 0 Architecture approved	Stakeholder signature	☐
1.2	Data Architecture approved	Stakeholder signature	☐
1.3	API Standards approved	Stakeholder signature	☐
1.4	Security Architecture approved	Stakeholder signature	☐
1.5	DevOps Architecture approved	Stakeholder signature	☐
1.6			☐

#	Item	Verified By	Status
	Performance Standards approved	Stakeholder signature	
1.7	Disaster Recovery approved	Stakeholder signature	☐
1.8	Schema Review Report produced	Document exists	☐
1.9	Stock Ledger Design produced	Document exists	☐
1.10	All ADRs (Architecture Decision Records) written	ADR folder populated	☐
1.11	Runbooks for top 10 ops tasks written	Runbook folder	☐
1.12	README.md in every package	File inspection	☐

2. Foundation Framework (Phase 0 Code)

2.1 Project Structure

#	Item	Verified By	Status
2.1.1	Monorepo setup (Turborepo + pnpm)	<code>pnpm install</code> works	☐

#	Item	Verified By	Status
2.1.2	apps/web , apps/backend , apps/mobile folders	Folder structure	☐
2.1.3	packages/ shared , packages/ prisma , packages/ui folders	Folder structure	☐
2.1.4	ESLint config enforces layer boundaries	bun run lint passes	☐
2.1.5	TypeScript strict mode enabled	tsconfig.json inspection	☐
2.1.6	No any types (without justification)	ESLint rule	☐

2.2 Configuration Framework

#	Item	Verified By	Status
2.2.1	Env validation (zod schema)	App refuses to start on missing env	☐
2.2.2	Feature flags implemented	features.isEnabled() works	☐
2.2.3	Config sources prioritized (env > DB > defaults)	Tested	☐

#	Item	Verified By	Status
2.2.4	Secrets manager integration	Secrets loaded at boot	☐

2.3 Error Framework

#	Item	Verified By	Status
2.3.1	Error hierarchy (BaseError + 10 subclasses)	Code inspection	☐
2.3.2	Error code catalog	packages/ shared export	☐
2.3.3	Error handler middleware	Errors shaped correctly	☐
2.3.4	5xx errors published to Sentry	Sentry integration	☐

2.4 Response Envelope

#	Item	Verified By	Status
2.4.1	Success envelope {success, data, meta}	All endpoints return this	☐
2.4.2	Error envelope {success: false, error, meta}	All errors return this	☐

#	Item	Verified By	Status
2.4.3	Pagination meta	List endpoints include meta	☐
2.4.4	Correlation ID in every response	X-Request-Id header	☐

2.5 Logging Framework

#	Item	Verified By	Status
2.5.1	Structured JSON logging (Pino)	Log format verified	☐
2.5.2	Request context (correlation ID, user, tenant)	Logs include context	☐
2.5.3	Sensitive field redaction	Logs don't contain secrets	☐
2.5.4	Log levels (fatal, error, warn, info, debug)	Configurable per env	☐

2.6 Validation Framework

#	Item	Verified By	Status
2.6.1	Zod schemas for every endpoint	Schema files exist	☐
2.6.2	Validation middleware	Invalid input → 400 with details	☐
2.6.3	Schema registry	Schemas registered at boot	☐

#	Item	Verified By	Status
2.6.4	Reusable schema fragments	Shared schemas in <code>packages/shared</code>	☐
2.6.5	OpenAPI spec generated from schemas	<code>openapi.yaml</code> exists	☐

2.7 Audit Framework

#	Item	Verified By	Status
2.7.1	<code>audit_logs</code> table migrated	DB inspection	☐
2.7.2	Audit log table is append-only	RLS policy enforced	☐
2.7.3	Audit middleware records mutations	All POST/PATCH/DELETE audited	☐
2.7.4	Workflow transitions audit logged	Transitions produce audit entry	☐
2.7.5	Audit query API (<code>/api/v1/audit/...</code>)	Endpoints work	☐
2.7.6	<code><AuditTrail / ></code> frontend component	Renders audit timeline	☐
2.7.7			☐

#	Item	Verified By	Status
	Hash chain for audit log integrity	Daily hash published externally	

2.8 Event Bus

#	Item	Verified By	Status
2.8.1	EventBus interface	Code inspection	☐
2.8.2	Outbox pattern implemented	event_outbox table + publisher job	☐
2.8.3	Event handler base class	BaseEventHandler exists	☐
2.8.4	Event catalog (all event names)	packages/shared export	☐
2.8.5	At-least-once delivery	Idempotency tested	☐
2.8.6	Event replay API	Admin can replay events	☐

2.9 Notification Engine

#	Item	Verified By	Status
2.9.1	Notification rule engine	Rules configurable per tenant	☐
2.9.2	In-app channel	Notifications table populated	☐

#	Item	Verified By	Status
2.9.3	Email channel	Test email sent via SMTP	☐
2.9.4	SMS channel (Twilio)	Stub or working	☐
2.9.5	WhatsApp channel	Stub or working	☐
2.9.6	Durable delivery (outbox pattern)	notification_outbox table	☐
2.9.7	Template system (Handlebars)	Templates versioned	☐
2.9.8	Subscription service	User can unsubscribe	☐
2.9.9	Delivery status tracked	Status visible in UI	☐

2.10 Workflow Engine

#	Item	Verified By	Status
2.10.1	BaseWorkflow class	Code inspection	☐
2.10.2	State machine with guards	Invalid transition rejected	☐
2.10.3	Pre/post transition hooks	Hooks fire on transition	☐
2.10.4	Transitions publish events	Events emitted	☐

#	Item	Verified By	Status
2.10.5	Transitions audit logged	Audit entries created	☐
2.10.6	Workflow registry	All workflows registered at boot	☐
2.10.7	Workflow inspection API	<code>/api/v1/workflows/:entityType/:id</code> works	☐
2.10.8	<code><WorkflowActions / ></code> frontend component	UI shows available actions	☐

2.11 Permission Framework

#	Item	Verified By	Status
2.11.1	RBAC middleware	Endpoints protected	☐
2.11.2	<code>@requirePermission</code> decorator	Used on routes	☐
2.11.3	Permission registry	All permissions catalogued	☐
2.11.4	3-layer enforcement (route, service, field)	Tested	☐
2.11.5	<code><PermissionGuard></code> frontend component	UI hides unauthorized	☐
2.11.6	Default roles seeded	<code>tenants</code> , <code>roles</code> , <code>permissions</code> seeded	☐

2.12 Multi-Tenancy

#	Item	Verified By	Status
2.12.1	<code>tenantId</code> on every business table	Schema inspection	☐
2.12.2	Prisma extension auto-scopes queries	Cross-tenant test blocked	☐
2.12.3	PostgreSQL RLS enabled	DB policy check	☐
2.12.4	TenantMiddleware loads context	AsyncLocalStorage populated	☐
2.12.5	Cross-tenant access audited at CRITICAL	Audit log check	☐

2.13 Transaction Helper

#	Item	Verified By	Status
2.13.1	<code>transaction()</code> helper	Code inspection	☐
2.13.2	Retry on deadlock/serialization	Retry test passes	☐
2.13.3	Long transaction detection (>5s warning)	Log warning fires	☐
2.13.4	Outbox pattern in transactions		☐

#	Item	Verified By	Status
		Events published after commit	

2.14 File Upload Service

#	Item	Verified By	Status
2.14.1	FileService interface	Code inspection	☐
2.14.2	S3 driver	Files uploaded to S3	☐
2.14.3	Local driver (dev)	Files saved locally	☐
2.14.4	MIME type whitelist	Invalid MIME rejected	☐
2.14.5	File size limits	Large file rejected	☐
2.14.6	Signed URLs (15-min TTL)	URLs expire	☐
2.14.7	Virus scan (ClamAV)	Infected file rejected	☐
2.14.8	File metadata in <code>file_uploads</code> table	DB inspection	☐

2.15 Background Job Framework

#	Item	Verified By	Status
2.15.1	BullMQ integration	Queue works	☐
2.15.2	Scheduled jobs (cron)	Jobs fire on schedule	☐

#	Item	Verified By	Status
2.15.3	Queued jobs	<code>jobs.enqueue()</code> works	☐
2.15.4	Retry with backoff	Failed job retried	☐
2.15.5	Dead-letter queue	Failed jobs land in DLQ	☐
2.15.6	Job observability API	<code>/api/v1/_internal/jobs</code> works	☐
2.15.7	Distributed lock (single instance per tenant)	Duplicate job prevented	☐

2.16 Base Classes

#	Item	Verified By	Status
2.16.1	<code>BaseRepository</code> with tenant scope, soft delete, optimistic lock	Code inspection	☐
2.16.2	<code>BaseService</code> with hooks + validation contract	Code inspection	☐
2.16.3	<code>BaseController</code> with CRUD handlers	Code inspection	☐
2.16.4	<code>BaseWorkflow</code> with state machine	Code inspection	☐
2.16.5	<code>BaseEventHandler</code>	Code inspection	☐

#	Item	Verified By	Status
2.16.6	BaseJob	Code inspection	☐
2.16.7	BaseMiddleware	Code inspection	☐

2.17 Frontend Foundation

#	Item	Verified By	Status
2.17.1	Axios client with interceptors	Code inspection	☐
2.17.2	React Query client + query key factory	Code inspection	☐
2.17.3	Auth context + permission guard	Login works, guards hide UI	☐
2.17.4	Shared UI components (DataTable, EntityForm, StatusBadge, etc.)	Component library exists	☐
2.17.5	Loading/error/empty states	All data views have states	☐
2.17.6	Toast notifications	Mutations show toasts	☐

3. Database

3.1 Database Setup

#	Item	Verified By	Status
3.1.1	PostgreSQL 16+ provisioned (production)	Live DB connection	☐
3.1.2	PostgreSQL 16+ provisioned (staging)	Live DB connection	☐
3.1.3	PostgreSQL 16+ in Docker (development)	docker-compose up works	☐
3.1.4	Connection pooling (PgBouncer/Supervisor)	Pool config verified	☐
3.1.5	Read replica provisioned	Replica accessible	☐
3.1.6	TLS required for all connections	Connection test	☐

3.2 Schema

#	Item	Verified By	Status
3.2.1	All Prisma models have reserved columns (id, tenant_id,	Schema inspection	☐

#	Item	Verified By	Status
	version, created_at, etc.)		
3.2.2	UUID v7 primary keys	Schema inspection	☐
3.2.3	Soft delete columns (deleted_at, deleted_by)	Schema inspection	☐
3.2.4	Optimistic locking (version field)	Schema inspection	☐
3.2.5	Foreign key indexes on every FK	DB inspection	☐
3.2.6	Partial indexes for soft-deleted tables	DB inspection	☐
3.2.7	Row-Level Security enabled	RLS policy check	☐
3.2.8	Check constraints for business rules	DB inspection	☐
3.2.9	Audit log table append-only	RLS policy	☐

3.3 Migrations

#	Item	Verified By	Status
3.3.1		DB inspection	☐

#	Item	Verified By	Status
	Initial migration applied to production		
3.3.2	Migration files in prisma/migrations/	Folder inspection	☐
3.3.3	Every migration has rollback SQL	Rollback tested	☐
3.3.4	Migrations forward-compatible (no downtime)	Tested in staging	☐
3.3.5	Migrations use CONCURRENTLY for new indexes	SQL inspection	☐
3.3.6	Seed scripts for reference data	Seed runs	☐
3.3.7	Seed scripts for default roles + permissions	Seed runs	☐

3.4 Backups

#	Item	Verified By	Status
3.4.1	PITR enabled (14-day retention)	Backup config	☐
3.4.2	Daily logical backup to S3	Backup job runs	☐

#	Item	Verified By	Status
3.4.3	Weekly logical backup (12-month retention)	Backup config	☐
3.4.4	Monthly logical backup to Glacier (7-year retention)	Backup config	☐
3.4.5	Backup encryption (SSE-KMS)	Encryption verified	☐
3.4.6	Daily restore-test job	Job runs, alerts on fail	☐
3.4.7	Weekly full restore to staging	Restore verified	☐
3.4.8	Cross-region backup replication	Replication verified	☐

3.5 Observability

#	Item	Verified By	Status
3.5.1	<code>pg_stat_statements</code> enabled	Extension check	☐
3.5.2	Slow query log (>500ms)	Log inspection	☐
3.5.3	Prometheus metrics exporter	Metrics scraped	☐
3.5.4	Database dashboard in Grafana	Dashboard exists	☐
3.5.5	Lock wait alerting	Alert fires on lock	☐

#	Item	Verified By	Status
3.5.6	Connection pool monitoring	Dashboard shows pool	☐

4. API

4.1 API Conformance

#	Item	Verified By	Status
4.1.1	All endpoints under <code>/api/v1/</code>	Route inspection	☐
4.1.2	REST conventions (verbs, status codes)	API test	☐
4.1.3	Plural kebab-case URLs	Route inspection	☐
4.1.4	UUID v7 IDs in URLs	Route test	☐
4.1.5	Standard response envelope	API test	☐
4.1.6	Error codes namespaced	Error response test	☐
4.1.7	Pagination (offset + cursor)	List endpoint test	☐
4.1.8	Filtering (operator suffixes)	Filter test	☐
4.1.9	Sorting (<code>-field</code> for desc)	Sort test	☐
4.1.10	Bulk operations (max 100)	Bulk endpoint test	☐
4.1.11	Idempotency key support	Idempotency test	☐
4.1.12	Rate limiting (4 response headers)	Rate limit test	☐
4.1.13	CORS configured	CORS test	☐
4.1.14	ETag + conditional requests	Cache test	☐

#	Item	Verified By	Status
4.1.15	gzip/Brotli compression	Compression test	☐

4.2 Authentication

#	Item	Verified By	Status
4.2.1	JWT auth on all endpoints (except login, health)	Unauth request → 401	☐
4.2.2	Argon2id password hashing	Hash inspection	☐
4.2.3	15-min access token TTL	Token inspection	☐
4.2.4	30-day refresh token (httpOnly cookie)	Cookie inspection	☐
4.2.5	Refresh token rotation	Old token rejected	☐
4.2.6	Refresh token replay detection	Replay → session revoked	☐
4.2.7	MFA (TOTP) for admin/finance/recall	MFA flow tested	☐
4.2.8	Account lockout (5 attempts → 30-min)	Lockout test	☐
4.2.9		Reset flow tested	☐

#	Item	Verified By	Status
	Password reset (token-based, 30-min TTL)		
4.2.10	JWT blocklist for compromised tokens	Blocklist test	☐

4.3 Authorization

#	Item	Verified By	Status
4.3.1	RBAC middleware on every endpoint	Permission test	☐
4.3.2	Row-level checks (user can only see own data)	Cross-user test	☐
4.3.3	Field-level response shaping	Sensitive field hidden	☐
4.3.4	Cross-tenant access blocked	Cross-tenant test	☐
4.3.5	Sensitive endpoints require MFA re-auth	Re-auth test	☐
4.3.6	All auth failures audit logged	Audit log check	☐

4.4 API Documentation

#	Item	Verified By	Status
4.4.1	OpenAPI 3.1 spec generated	openapi.yaml exists	☐
4.4.2	Every endpoint documented	Spec inspection	☐
4.4.3	TypeScript SDK auto-generated	@suop/api-client package	☐
4.4.4	API changelog maintained	CHANGELOG.md updated	☐
4.4.5	Swagger UI (dev only)	/api/docs works in dev	☐

5. Security

5.1 Transport Security

#	Item	Verified By	Status
5.1.1	TLS 1.3 enforced	SSL Labs A+ grade	☐
5.1.2	HSTS header (2-year, includeSubDomains, preload)	Header inspection	☐
5.1.3	Auto-renewing certificates	Renewal verified	☐
5.1.4	DB connection over TLS	Connection test	☐

#	Item	Verified By	Status
5.1.5	Redis connection over TLS	Connection test	☐

5.2 Application Security

#	Item	Verified By	Status
5.2.1	All inputs validated by zod schema	Validation test	☐
5.2.2	SQL injection prevention (Prisma parameterized)	SAST scan	☐
5.2.3	XSS prevention (React auto-encoding, no dangerouslySetInnerHTML)	Code scan	☐
5.2.4	CSRF protection (SameSite cookies)	Cookie test	☐
5.2.5	SSRF prevention (URL allowlist + IP blocking)	Outbound URL test	☐
5.2.6	Content Security Policy header	Header inspection	☐
5.2.7	Security headers (nosniff, frame-deny, etc.)	Header inspection	☐
5.2.8	Sensitive column encryption (pgcrypto)	DB inspection	☐
5.2.9	Password pepper from secrets manager	Config inspection	☐
5.2.10	No secrets in code/git	Secret scan	☐

5.3 Infrastructure Security

#	Item	Verified By	Status
5.3.1	VPC with private subnets	Network config	☐
5.3.2	DB in isolated subnet (no internet)	Network config	☐
5.3.3	Security groups least privilege	SG inspection	☐
5.3.4	S3 buckets private (no public access)	Bucket policy	☐
5.3.5	S3 versioning enabled	Bucket config	☐
5.3.6	S3 cross-region replication	Replication check	☐
5.3.7	Container non-root user	Dockerfile inspection	☐
5.3.8	Container read-only filesystem	Container test	☐
5.3.9	Image scanning (Trivy)	CI scan passes	☐
5.3.10	SBOM generated per image	SBOM artifact	☐

5.4 Security Testing

#	Item	Verified By	Status
5.4.1	SAST on every PR (SonarQube)	CI passes	☐
5.4.2	Dependency scanning (Snyk + npm audit)	CI passes	☐
5.4.3	Secret scanning (git-secrets)	CI passes	☐
5.4.4	Container image scanning (Trivy)	CI passes	☐
5.4.5	Annual penetration test		☐

#	Item	Verified By	Status
		Pen test report	
5.4.6	OWASP Top 10 coverage verified	Coverage doc	☐

5.5 Audit & Compliance

#	Item	Verified By	Status
5.5.1	All mutations audit logged	Audit log inspection	☐
5.5.2	Audit log immutable (RLS)	UPDATE/DELETE rejected	☐
5.5.3	Audit log hash chain	Hash verified	☐
5.5.4	Audit log retention (7 years)	Retention config	☐
5.5.5	FSSAI compliance (2-year records)	Retention verified	☐
5.5.6	GST compliance (6-year records)	Retention verified	☐
5.5.7	DPDP compliance (DSAR workflow)	DSAR test	☐

6. DevOps

6.1 CI/CD

#	Item	Verified By	Status
6.1.1	CI pipeline (lint, typecheck, test, build)	CI runs on PR	☐

#	Item	Verified By	Status
6.1.2	CD pipeline (staging auto, production manual)	Deploy verified	☐
6.1.3	Blue-green deployment	Zero-downtime deploy	☐
6.1.4	Auto-rollback on smoke test failure	Rollback tested	☐
6.1.5	Migration tested in CI	Migration runs in CI	☐
6.1.6	All gates blocking	PR blocked on fail	☐

6.2 Docker

#	Item	Verified By	Status
6.2.1	Dockerfile.backend (multi-stage)	Build works	☐
6.2.2	Dockerfile.web (multi-stage, Next.js standalone)	Build works	☐
6.2.3	docker-compose.yml (full stack)	docker-compose up works	☐
6.2.4	Non-root user in containers	Container test	☐
6.2.5			☐

#	Item	Verified By	Status
	Health check in Dockerfile	Health endpoint works	

6.3 Infrastructure as Code

#	Item	Verified By	Status
6.3.1	Terraform for all infrastructure	terraform plan works	☐
6.3.2	State in S3 backend with lock	State config	☐
6.3.3	Drift detection daily	Drift alert works	☐
6.3.4	Staging + production environments	Both deployable	☐

6.4 Git Strategy

#	Item	Verified By	Status
6.4.1	Branch protection on main + develop	Branch settings	☐
6.4.2	Signed commits required	Commit signatures	☐
6.4.3	PR review (1 for code, 2 for security)	PR template	☐
6.4.4	Squash & merge	Merge history	☐
6.4.5	Conventional commits enforced	Commit lint	☐

7. Performance

7.1 API Performance

#	Item	Verified By	Status
7.1.1	p95 latency targets met (per endpoint type)	Load test	☐
7.1.2	No N+1 queries	Query count test	☐
7.1.3	Cursor pagination for >100k row tables	API test	☐
7.1.4	Page size max 200	API test	☐
7.1.5	ETag on GET responses	Header test	☐
7.1.6	gzip/Brotli enabled	Compression test	☐

7.2 Database Performance

#	Item	Verified By	Status
7.2.1	Every FK has an index	DB inspection	☐
7.2.2	EXPLAIN ANALYZE shows no seq scans >10k rows	Plan inspection	☐
7.2.3	Slow query log enabled (>500ms)	Log config	☐
7.2.4		Config inspection	☐

#	Item	Verified By	Status
	Auto-vacuum tuned		
7.2.5	Partitioning for large tables (audit, ledger)	Partition check	☐
7.2.6	Cache hit ratio >95%	pg_stat_database	☐

7.3 Frontend Performance

#	Item	Verified By	Status
7.3.1	Lighthouse Performance ≥ 90	Lighthouse run	☐
7.3.2	LCP <2.5s	Lighthouse	☐
7.3.3	FID <100ms	Lighthouse	☐
7.3.4	CLS <0.1	Lighthouse	☐
7.3.5	Initial bundle <300KB gzipped	Bundle analyzer	☐
7.3.6	Per-route bundle <100KB gzipped	Bundle analyzer	☐
7.3.7	Code splitting per route	Bundle inspection	☐
7.3.8	Image optimization (Next.js Image)	Code inspection	☐

7.4 Caching

#	Item	Verified By	Status
7.4.1	Redis cache operational	Cache hit/miss metrics	☐
7.4.2	Reference data cached (1h TTL)	Cache test	☐

#	Item	Verified By	Status
7.4.3	Master data cached (5min TTL)	Cache test	☐
7.4.4	Dashboard aggregations cached (5min TTL)	Cache test	☐
7.4.5	Cache hit rate >80%	Redis metrics	☐
7.4.6	Cache invalidation on update	Update → cache miss	☐
7.4.7	CDN configured (Cloudflare)	CDN test	☐
7.4.8	Static assets cached (1 year)	CDN config	☐

7.5 Load Testing

#	Item	Verified By	Status
7.5.1	Normal load test (100 users) passes	k6 run	☐
7.5.2	Peak load test (500 users) passes	k6 run	☐
7.5.3	Stress test (find breaking point) documented	Report exists	☐
7.5.4	Spike test passes	k6 run	☐
7.5.5	Soak test (8 hours) passes	k6 run	☐
7.5.6		CI passes	☐

#	Item	Verified By	Status
	Performance regression tests in CI		

8. Disaster Recovery

8.1 Backup & Recovery

#	Item	Verified By	Status
8.1.1	Daily restore-test job	Job runs, alerts work	☐
8.1.2	Weekly full restore to staging	Restore verified	☐
8.1.3	Monthly DR failover drill	Drill report	☐
8.1.4	Annual full disaster simulation	Simulation report	☐
8.1.5	Glacier restore tested	Restore time documented	☐

8.2 High Availability

#	Item	Verified By	Status
8.2.1	Multi-AZ DB (primary + 2 replicas)	DB config	☐
8.2.2	Auto-failover for DB	Failover test	☐
8.2.3	Cross-region async replica	Replication lag <1min	☐
8.2.4	Multi-AZ backend (min 2 tasks)	ECS config	☐

#	Item	Verified By	Status
8.2.5	Multi-AZ load balancer	ALB config	☐
8.2.6	Route53 health check + failover	DNS config	☐

8.3 DR Documentation

#	Item	Verified By	Status
8.3.1	15 DR runbooks written	Runbook folder	☐
8.3.2	DR contact tree documented	Doc inspection	☐
8.3.3	BCP manual workaround procedures	BCP doc	☐
8.3.4	Communication templates drafted	Template folder	☐
8.3.5	Regulatory notification procedures	Doc inspection	☐

8.4 DR Testing

#	Item	Verified By	Status
8.4.1	Tabletop exercise (quarterly)	Exercise report	☐
8.4.2	DR failover drill (monthly)	Drill report	☐
8.4.3	Full restore test (weekly)	Test report	☐
8.4.4	Full BCP test (annual)	Test report	☐
8.4.5	Full disaster simulation (annual)	Simulation report	☐

9. Monitoring & Observability

9.1 Metrics

#	Item	Verified By	Status
9.1.1	Prometheus scraping backend	Metrics endpoint	☐
9.1.2	Grafana dashboards (system, DB, business)	Dashboards exist	☐
9.1.3	Database metrics (connections, latency, locks)	Dashboard	☐
9.1.4	Redis metrics (memory, hit rate)	Dashboard	☐
9.1.5	Business metrics (POs/day, GRNs/day, NCRs/day)	Dashboard	☐

9.2 Logs

#	Item	Verified By	Status
9.2.1	Structured JSON logs (Pino)	Log format	☐
9.2.2	Log aggregation (Loki or ELK)	Logs searchable	☐

#	Item	Verified By	Status
9.2.3	Sensitive field redaction	Log inspection	☐
9.2.4	Log retention (30 days hot, 90 days warm)	Retention config	☐
9.2.5	Correlation ID in every log line	Log inspection	☐

9.3 Tracing

#	Item	Verified By	Status
9.3.1	OpenTelemetry instrumentation	Trace export	☐
9.3.2	Traces exported to Tempo/Jaeger	Trace inspection	☐
9.3.3	100% error traces captured	Trace test	☐
9.3.4	10% normal traces sampled	Sampling config	☐

9.4 Alerting

#	Item	Verified By	Status
9.4.1	PagerDuty integration (critical alerts)	Test page	☐
9.4.2	Slack integration (warning alerts)	Test alert	☐
9.4.3	High error rate alert (>5% for 5min)	Alert test	☐

#	Item	Verified By	Status
9.4.4	Service down alert	Alert test	☐
9.4.5	DB down alert	Alert test	☐
9.4.6	High latency alert (p95 >1s for 10min)	Alert test	☐
9.4.7	Disk space alert (>80%)	Alert test	☐
9.4.8	Business alerts (NCR critical, recall, CAPA overdue)	Alert test	☐
9.4.9	Security alerts (failed login spike, permission denied spike)	Alert test	☐
9.4.10	Alert fatigue prevention (grouping, inhibition)	Config verified	☐

9.5 Error Tracking

#	Item	Verified By	Status
9.5.1	Sentry integration	Errors captured	☐
9.5.2	Source maps uploaded	Stack traces readable	☐
9.5.3	Release tracking	Releases tagged in Sentry	☐

#	Item	Verified By	Status
9.5.4	Frontend errors captured	Error test	☐

9.6 Uptime Monitoring

#	Item	Verified By	Status
9.6.1	External uptime monitor (Pingdom/UptimeRobot)	Monitor configured	☐
9.6.2	Synthetic transaction monitoring (Checkly)	Synthetics run	☐
9.6.3	Status page (public)	<code>status.sudhamrit.com</code> live	☐
9.6.4	Real User Monitoring (Sentry Performance)	RUM data flowing	☐

10. Testing

10.1 Test Infrastructure

#	Item	Verified By	Status
10.1.1	Vitest configured	<code>bun run test:unit</code> works	☐
10.1.2	Testcontainers configured	Integration tests run	☐
10.1.3	Playwright configured	E2E tests run	☐

#	Item	Verified By	Status
10.1.4	k6 configured	Load tests run	☐
10.1.5	Test database isolated	Tests don't pollute	☐

10.2 Test Coverage

#	Item	Verified By	Status
10.2.1	Foundation code coverage $\geq 90\%$	Coverage report	☐
10.2.2	Business code coverage $\geq 70\%$	Coverage report	☐
10.2.3	Coverage gate in CI	CI fails below threshold	☐
10.2.4	Mutation tests (Stryker) on foundation	Mutation score $\geq 80\%$	☐

10.3 Test Types

#	Item	Verified By	Status
10.3.1	Unit tests for every service method	Test count	☐
10.3.2	Unit tests for every workflow transition	Test count	☐
10.3.3	Integration tests for every endpoint	Test count	☐
10.3.4	E2E tests for top 20 user journeys	Test count	☐
10.3.5	Performance regression tests	Nightly run	☐

#	Item	Verified By	Status
10.3.6	Contract tests (frontend ↔ backend)	CI passes	☐
10.3.7	Smoke tests post-deploy	Deploy triggers tests	☐

11. First Business Journey (Procure-to-Receive)

11.1 Organization Module

#	Item	Verified By	Status
11.1.1	Schema migrated	DB inspection	☐
11.1.2	Repository implemented	Code inspection	☐
11.1.3	Service with business rules	Code inspection	☐
11.1.4	Workflow (Plant lifecycle)	State machine test	☐
11.1.5	API endpoints (CRUD + workflow)	API tests	☐
11.1.6	Frontend wired to API	UI test	☐
11.1.7	RBAC enforced	Permission test	☐
11.1.8	Audit logging	Audit log check	☐
11.1.9	Notifications (plant activation)	Notification test	☐
11.1.10	Unit + integration tests	Tests pass	☐

11.2 Auth & RBAC Module

#	Item	Verified By	Status
11.2.1	Schema migrated	DB inspection	☐
11.2.2	Real auth (Argon2id + JWT)	Auth test	☐
11.2.3	MFA (TOTP)	MFA test	☐
11.2.4	Refresh token rotation	Rotation test	☐
11.2.5	User lifecycle state machine	Workflow test	☐
11.2.6	RBAC middleware on every endpoint	Permission test	☐
11.2.7	Frontend login wired to real API	Login test	☐
11.2.8	RBAC UI (roles, permissions) wired	UI test	☐
11.2.9	Audit logging (login, logout, permission changes)	Audit check	☐
11.2.10	Notifications (welcome, password reset)	Notification test	☐
11.2.11		Coverage report	☐

12. Production Launch Readiness

12.1 Final Pre-Launch

#	Item	Verified By	Status
12.1.1	Production environment provisioned	Terraform applied	☐
12.1.2	DNS configured	DNS lookup works	☐
12.1.3	TLS certificate valid	SSL Labs test	☐
12.1.4	All env vars set in secrets manager	Config check	☐
12.1.5	Database migrated	Migration applied	☐
12.1.6	Seed data loaded (roles, permissions, reference data)	Seed runs	☐
12.1.7	Initial tenant + admin user created	Login test	☐
12.1.8	Smoke tests pass on production	Test run	☐
12.1.9	Monitoring alerts firing	Alert test	☐

#	Item	Verified By	Status
12.1.10	Status page live	Page accessible	☐
12.1.11	Backup jobs scheduled	Jobs run	☐
12.1.12	DR drill completed	Drill report	☐
12.1.13	On-call rotation set up	PagerDuty schedule	☐
12.1.14	Customer support trained	Training complete	☐
12.1.15	Documentation for end users	User guide exists	☐

12.2 Sign-off

Role	Name	Signature	Date
Project Owner			
Lead Architect			
Engineering Lead			
Operations Lead			
Security Lead			
QA Lead			

Final Production Readiness Score

Calculation: - Total checkboxes: ~350 - Each checkbox worth ~0.285% -
 Production Ready % = (checked / total) × 100

Status: - ☐ 0% — Documentation phase - ☐ 25% — Foundation framework complete - ☐ 50% — First business journey complete - ☐ 75% — All modules wired - ☐ 90% — Testing + monitoring complete - ☐ 100% — Production ready (sign-off required)

Target: 100% before any code is “done”

Approval Block

By approving this checklist, the Project Owner authorizes:

- 1. **Permanent freeze** of all architecture documentation
- 2. **No additional documentation** unless explicitly requested
- 3. **All future responses contain production code only**
- 4. Implementation begins with Phase 0 Foundation code

Approved by: _____ **Date:** _____

Signature: _____

Once approved, the documentation phase is permanently frozen. Implementation begins.

End of Final Production Readiness Checklist